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SUPERSEDING
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MILITARY SPECIFICATION

LENS ASSEMBLY, EYEPIECE, AN/PVS-4 AND AN/TVS-5

This specification is approved for use by US Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Lens Assembly, Eyepiece, AN/PVS-4 and AN/TVS-5 (see 6.1), which is referred to herein as the eyepiece lens.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2.e).

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command ATTN: AMSEL-ED-TM, Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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SPECIFICATIONS

MILITARY

MIL-O-13830	-	Optical Component for Fire Control Instrument; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-P-11268	-	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	-	Marking of Electronic Items

STANDARDS

MILITARY

MIL-STD-150	-	Photographic Lenses
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg #4, Section D, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.e).

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-850401	-	Lens Assembly, Eyepiece, AN/PVS-4 and AN/TVS-5

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The Lens Assembly Eyepiece is part of the Night Vision Sight, Individual Served Weapon, AN/PVS-4 and Night Vision Sight, Crew Served Weapon, AN/TVS-5.

3.2 Qualification. Unless otherwise specified by the contract or purchase order, Lens Assemblies furnished under this specification shall be products approved as qualified products (see 3.10.4 and 6.4).

3.2.1 Initial production testing. When specified in the contract or purchase order, the contractor shall furnish IPT eyepiece lenses in accordance with 4.3.

3.3 Construction. The eyepiece lens shall be constructed in accordance with SM-D-850401 and all provisions of this specification as specified herein. Unless otherwise specified, surface treatment and finish shall conform to MIL-STD-171.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Material not specified shall be selected by the contractor, shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Components. The eyepiece lens shall consist of optical components assembled within a housing as specified on the drawings.

3.6 Performance characteristics.

3.6.1 Effective focal length. The effective focal length shall be 26.5, ± 0.3mm.

3.6.2 Flange focal distance. The flange focal distance of the eyepiece at zero diopter shall be 4.753 ± 0.207, -0.435mm.

3.6.3 Field curvature. The field curvature of the eyepiece lens shall deviate no more than ± 1/4 diopter over the full 25mm flat format.

3.6.4 Linear distortion. The linear distortion of the eyepiece lens shall be between 0 and 9 percent pincushion distortion at the edge of the 25mm format.

3.6.5 Exit pupil. The exit pupil diameter shall be 5.0, ± 0.2mm at eye relief of 28.0, ± 0.5mm.

3.6.6 Transmission. The transmission of the eyepiece shall be not less than 80 percent over the full eyepiece aperture for the spectral output of the P-20 phosphor.

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3.6.7 Modulation transfer function. The modulation transfer function (MTF) of the eyepiece lens shall not be less than that specified in Figure 1.

3.6.8 Diopter adjustment. The diopter adjustment shall operate smoothly without binding throughout its complete range movement at all operating temperatures specified herein. Dynamic torque shall be between 2 and 20 inch-pounds.

3.6.9 Diopter range. The eyepiece focus range shall not be less than 8 diopters when one rotational stop is set at -6 (+1/4 -0) diopters.

3.7 Environmental.

3.7.1 Vibration. The eyepiece lens shall not be damaged (see 3.10.1) by simple harmonic motion having an amplitude of 0.015 inch, (0.03 inch total excursion) with the frequency being varied between 5 and 55 hertz (Hz).

3.7.2 High temperature storage and operation. The eyepiece lens shall not be damaged (see 3.10.1) by storage in any temperature from +23° to +68°C and shall operate as specified herein at any temperature from +23° to +52°C.

3.7.3 Low temperature storage and operation. The eyepiece lens shall not be damaged (see 3.10.1) by storage at any temperature from +23° to -51°C and shall operate as specified herein at any temperature from +23° to -51°C.

3.7.4 Altitude. The eyepiece lens shall not be damaged (see 3.10.1) when operated at a pressure equivalent to 10,000 feet altitude above sea level.

3.7.5 Temperature shock. The eyepiece lens shall not be damaged (see 3.10.1) after being subjected to temperature changes between +23° and +68°C in 5 minutes and between +23° and -51°C in 5 minutes.

3.7.6. Humidity. The eyepiece lens shall not be damaged (see 3.10.1) after storage up to +68°C minimum with 95, ± 5 percent relative humidity.

3.7.7 Immersion. The eyepiece lens shall not be damaged (see 3.10.1) after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes.

3.7.8 High intensity shock. The eyepiece lens shall not be damaged (see 3.10.1) by a sequence of shocks applied in each direction along each of 3 mutually perpendicular axes (horizontal, vertical, and optical axis when the reticle pattern is erect). The shocks shall be half sine pulses and shall have a time duration of 4 milliseconds, ± 5 percent. Shock pulses applied along the axes horizontal and perpendicular to the optical axis shall have a peak amplitude of 50g's, ± 15 percent (see 3.10.2). Shock pulses applied along the optical axis shall have a peak amplitude of 100, ± 15g's.

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3.8 Marking. Unless otherwise specified, the eyepiece lens and all components shall be marked per MIL-M-13231.

3.9 Workmanship. All eyepieces shall be free from cracks, splits, cold flow, shrinkage, inclusions, porosity, or any similar characteristics. Threads shall be full and undamaged for the entire length or depth as required on the applicable drawing. Eyepieces shall be free from burrs and shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material. The cleaning methods used shall not damage any of the eyepieces, nor shall the parts be contaminated by the cleaning agent. Quality requirements defined as scratches, digs, edge chips, bubbles, coating defects and cementing defects shall be in accordance with MIL-O-13830 except that inspection of optical cleanliness shall be with the unaided eye (without 3X magnification). All moving parts and adjustments shall be examined to insure that they move freely throughout their entire range without sticking, binding, or creeping. The requirements of MIL-STD-454, Requirement 9, shall apply unless otherwise specified.

3.10 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.10.1 Damage. Breakage, loosening, shifting, evidence or corrosion or failure of any finish, hardware, connection or component; leakage or condensation of moisture within the eyepiece lens; or degradation in input or output characteristics.

3.10.2 "g". "g" is a unitless measure of the quantity of units of force associated with the earths gravity that a object is subjected to.

3.10.3 Room temperature. Room temperature is defined as +23°C, ± 3°C.

3.10.4 Qualified product. A product that has successfully met all the requirements of an IPT of First Article witnessed by the CECOM, Concurrent Engineering Directorate or its designated representative. Should the supplier make substantial changes in the process, personnel, location, or equipment used to produce a qualified product, or have not delivered the qualified product to the Federal Government within 12 months of contract award, and IPT shall be performed to the extent necessary to verify continued compliance to specification requirements for all characterisitcs of the product affected by the changes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet the requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Components and materials inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards as specified herein.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Initial production testing (4.3).
b. Quality conformance inspection (4.5).
c. Packaging inspection (4.8).

4.3 Initial production testing.

4.3.1 IPT samples. The quantity of initial production testing samples shall be specified in the contract or purchase order and shall be selected from the first production lot. Disposition of IPT samples shall be in accordance with the contract or purchase order (see 6.2.b).

4.3.2 Inspection. Each IPT eyepiece component shall be inspected in process in accordance with the defects listed in Table I and per the requirements of 3.3, 3.4, 3.8 and 3.9. Presence of one or more defects shall be cause for rejection of that eyepiece component.

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TABLE I. Inspections.

Defects
Performing requirements specified in notes appearing on drawings.
Linear or diametrical tolerances of 0.002 inch or less.
Angular dimension having a tolerance of 15 minutes of arc or less.
Machine finishes of 32 micro-inches RMS or finer.
Tolerances of position or form 0.002 inch or less.
Screw thread Class 3 or better.
Load requirements for springs.
Lubrication requirements including dry lubricant finishes.
Material hardness when specified on drawings.
Diopter control moves freely without binding.

4.3.3 Tests. Following successful completion of the inspections specified in 4.3.1, each IPT eyepiece lens shall be subjected to all inspections in Table II. Failure of any inspection shall be cause for rejection of that eyepiece lens. Inspections may be performed in any order.

TABLE II. IPT inspection.

Inspection	Requirement paragraph	Test paragraph
Effective focal length	3.6.1	4.6.1
Flange focal distance	3.6.2	4.6.2
Field curvature	3.6.3	4.6.3
Linear distortion	3.6.4	4.6.4
Exit pupil	3.6.5	4.6.5
Transmission	3.6.6	4.6.6
Modulation transfer function	3.6.7	4.6.7
Diopter adjustment	3.6.8	4.6.8
Diopter range	3.6.9	4.6.9
Vibration	3.7.1	4.7.1
High temperature storage and operation	3.7.2	4.7.2
Low temperature storage and operation	3.7.3	4.7.3
Altitude	3.7.4	4.7.4
Temperature shock	3.7.5	4.7.5
Humidity	3.7.6	4.7.6
Immersion	3.7.7	4.7.7
High intensity shock	3.7.8	4.7.8

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4.4 Inspection conditions. Unless otherwise specified in the contract or purchase order, inspections shall be performed as specified herein. Tolerances on specified illumination levels shall be ± 10 percent. Neutral density filters used in test equipment shall have transmission characteristics within 10 percent of the nominal filter transmission from 0.35 to 1.0 micrometer. When a collimator is used, effective focal length (EFL) of the collimator shall be at least six times the EFL of the eyepiece lens under test. The clear aperture of the collimator shall be six inches or greater. Tolerance on 2854K color temperature shall be ± 200K. References within requirements and tests to "initial measurements", "operate", "pretest", or "performance check during test" shall be defined as the modulation transfer function test.

4.5 Quality conformance inspection.

4.5.1 Group A inspection. Each eyepiece lens shall be inspected per 3.9, Table I and III. Failure of any eyepiece lens to pass all the requirements of Table I and III and 3.3, 3.4, 3.8 and 3.9 shall be cause for rejection of that eyepiece lens.

TABLE III. Group A Inspection.

Inspection	Requirement paragraph	Test paragraph
Exit pupil	3.6.5	4.6.5
Transmission	3.6.6	4.6.6
Modulation transfer function	3.6.7	4.6.7

4.5.2 Group B inspection. Eyepiece lenses which have passed the inspections specified in 4.5.1 shall be formed into lots by the contractor. Sample eyepiece lenses shall be selected and subjected to the inspection of Table IV. Sampling shall be in accordance with the contract or purchase order (see 6.2.c). Inspections shall be performed in any order.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Effective focal length	3.6.1	4.6.1
Flange focal distance	3.6.2	4.6.2
Field curvature	3.6.3	4.6.3
Linear distortion	3.6.4	4.6.4
Diopter adjustment	3.6.8	4.6.8
Diopter range	3.6.9	4.6.9

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4.5.3 Group C inspection. Group C inspections shall be conducted on eyepiece lenses selected from lots which have passed the inspections in 4.5.1. The sample(s) shall be subjected to the inspections specified in Table V. Sampling shall be as specified in accordance with the contract or purchase order (see 6.2.c). Inspections may be performed in any order except that in Subgroup 1, Immersion shall be last, and in Subgroup 2, Humidity shall be last.

TABLE V. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Vibration	3.7.1	4.7.1
High temperature storage and operation	3.7.2	4.7.2
Low temperature storage and operation	3.7.3	4.7.3
Immersion	3.7.7	4.7.7
Subgroup 2		
High intensity shock	3.7.8	4.7.8
Temperature shock	3.7.5	4.7.5
Altitude	3.7.4	4.7.4
Humidity	3.7.6	4.7.6

4.5.4 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order (see 6.2.d).

4.5.5 Disposition of Group C inspection units. Any eyepiece lens which have been subjected to Group C inspections shall be subjected (after refurbishment if necessary) to the test of Tables III and IV prior to acceptance. Failure of any test of Tables III and IV shall be cause to reject that eyepiece lens.

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4.6 Performance characteristics.

4.6.1 Effective focal length. Effective focal length of the eyepiece shall be measured with the eyepiece run backwards (i.e., collimated light enters the eye lens) with a 5.0mm, ± 0.2mm exit pupil at a 28.0, (± 0.5)mm eyerelief distance between the collimator and the eyepiece. This assembly is placed on a goniometric table that includes a measuring microscope. The light source is a collimated beam, with a pinhole or slit target, of sufficient bundle size as to fill the lens under test at all possible field angles. The size of the pinhole/slit width as geometrically projected through the lens under test shall not be greater than 0.005mm. The microscope measuring ways are previously aligned to the point source or slit source target.
The effective focal length (EFL) of the objective lens assembly is determined by measuring the image translation produced by rotating the goniometer through an angle of ± one (1) degree. The EFL is the quotient of the difference in the measuring microscope readings divided by two over the tangent of one degree. Failure to meet the requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Flange focal distance. The flange focal distance of the eyepiece shall be measured as part of the on-axis MTF test, 4.6.7.1. Failure to meet the requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Field curvature. Field curvature of the eyepiece shall be measured in accordance with MIL-STD-150, Method 16. A measuring microscope shall be used in place of a photographic plate. In order to use this procedure, the eyepiece shall be run backwards (i.e., collimated light enters the eye lens) with a 5.0mm, ± 0.2mm exit pupil at a 28.0, ± 0.5mm eye relief distance between the collimator and the eyepiece. Separate measurements shall be taken for radial and tangential targets. The surface of best focus shall be defined as the average between the radial and tangential surface. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Linear distortion. Using the set-up as described for the effective focal length test, measure the semi-field angle for a semi-field height of plus and minus twelve and one-half (12.5) millimeters. Using the average of the semi-field angle (sfa), the percent distortion is 100*(12.5-efl*tan(sfa))/(efl*tan(sfa)). Failure to meet the requirements of paragraph 3.6.4 shall constitute failure of this test.

4.6.5 Exit Pupil. Brightly illuminate a cross target at the infinity focal plane of a collimator. Position an opaque mask on the eyepiece at the 28.0, ± 0.5mm eye relief distance. The mask shall have two pinholes (1mm diameter) 4mm apart (center to center), centered on the optical axis, and aligned in the azimuth direction. Place the eyepiece on a nodal slide so that the mask is between the collimator and the eyepiece. Rotate the eyepiece in azimuth 27°, ± 1° in each direction and locate the image of the cross target using a traveling microscope of 20X or greater. Failure to observe the cross target through each of the pinholes at both +27° and -27° rotation from center shall constitute failure of this test.

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4.6.6 Transmission. Transmission of the eyepiece shall be measured in accordance with MIL-STD-150. Failure to comply with the requirements of 3.6.6 shall constitute failure of this test.

4.6.7 Modulation transfer function (MTF). The eyepiece lens shall be tested on an MTF apparatus that yields sine-wave MTF response. The MTF apparatus shall have a light source and detector whose combination shall yield a spectral response curve complying with Figure 2. (If a microscope objective is used to relay the image of the eyepiece lens, the numerical aperture (NA) of the microscope objective shall not be less than 0.45). The MTF apparatus shall allow the eyepiece to be tested either continuously or at a minimum of 4 distinct evenly distributed points from 0 to 40 lp/mm. The area analyzed shall be at least 0.5mm in diameter. The calibration of the MTF apparatus shall be performed before the test data is valid. The calibration procedure to be used is to measure the MTF of a known size target (e.g. slit or pinhole). All references to size are in the aerial image. The MTF of a pinhole is known mathematically as a first order Bessel Function. MTF = ($2J_1$ (x))/ x where x = 3.14156 * (lp/mm) * (pinhole diameter in mm). The MTF of a pinhole goes to zero at a spatial frequency 1.22 times its reciprocal diameter. The MTF of a slit is known mathematically as (sin (x))/(x), where x = 3.14156 * (lp/mm) * (slit width). The units for x is in radians. The MTF of a slit goes to zero at a spatial frequency at its reciprocal width. The procedure shall be to measure the MTF of the lens and then measure the MTF of the lens with a very large pinhole/slit. In the case of microprocessor based MTF apparatus which correct for the pinhole/slit size, the size of the pinhole/slit entered into the computer software, when measuring the lens with large pinhole/slit should not be greater than 1/40 of the larger pinhole/slit. The geometrical size of the aerial image shall be 0.030mm ± 0.001. The geometrical size of the slit shall be 0.025 ± 0.001. The MTF of the large pinhole/slit can be determined by taking the ratio of the MTF curves of the lens with large pinhole/slit, divided by the MTF of the lens under test. Differences in modulation of greater than three percent will invalidate the MTF results. See Figure 3 for MTF as a function of spatial frequency and pinhole/slit width. Failure to meet the requirements of 3.6.7. shall constitute failure of this test.

4.6.7.1 On-axis MTF. The eyepiece lens shall be measured on-axis with the lens in the normal operation orientation (reticle erect) and then rotated about the optical axis 90° and remeasured in the same focal plane. Failure of either measurement to meet requirements of 3.6.7 shall constitute failure of this test.

4.6.8 Diopter adjustment. Diopter adjustments shall be made throughout the specified operating temperature ranges. The dynamic torque shall be between 2 and 20 inch-pounds. Failure to meet the requirements of 3.6.8 shall constitute failure of this test.

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4.6.9 Diopter range. With a target located at the focal plane of the eyepiece lens, use a dioptometer to determine that the range of the eyepiece lens is -6, (+1/4, -0) diopter to +2 diopters. Failure to meet requirements of 3.6.9 shall constitute failure of this test.

4.7 Environmental. The eyepiece shall be retested to meet the requirements of 3.6.5, 3.6.6, and 3.6.7 after the completion of each environmental test. Failure to meet these requirements shall constitute failure due to degradation of input or output characteristics per the damage technical interpretation of 3.10.1. Operating tests required during environmental tests shall be limited to a visual inspection for damage (see 3.10.1). Temperature chambers shall maintain specified temperatures within ± 2.0°C.

4.7.1 Vibration. The eyepiece shall be subjected to Method 201A of MIL-STD-202. The induced vibratory force shall be applied in each of three (3) mutually perpendicular planes of which one (1) is perpendicular to the optical axis. The duration of vibration shall be five (5) minutes minimum in each plane. Failure to meet the requirements of 3.6.5, 3.6.6, and 3.7.1 shall constitute failure of this test.

4.7.2 High temperature storage and operation. The eyepiece shall be subjected to the test of Method 501.1, Procedures I, of MIL-STD-810, except that the highest storage temperature of Steps 2 and 3 shall be +68°C for a minimum period of six (6) hours. Steps 4 and 5 highest operating temperature shall be +52°C for a minimum of (3) hours. Step 6 stabilization period shall be a minimum of two (2) hours. Failure to meet the requirements of 3.6.5, 3.6.6, 3.6.7 and 3.7.2 shall constitute failure of this test.

4.7.3. Low temperature storage and operation. The eyepiece shall be subjected to the test of Method 502.1, Procedures I, of MIL-STD-810, except that the lowest storage temperature of Steps 2 shall be -51°C for a minimum of nine (9) hours. Step 3 shall be deleted. Step 4 and 5 lowest operating temperature shall be -51°C for a minimum of nine (9) hours. Steps 2 and 6 stabilization period shall be a minimum of two (2) hours. Failure to meet the requirements of 3.6.5, 3.6.6, 3.6.7, and 3.7.3 shall constitute failure of this test.

4.7.4 Altitude. The eyepiece shall be subjected to the test of Method 500.1, Procedure I, of MIL-STD-810, except that the minimum chamber pressure of Step 2 shall be effective to no less than 10,000 feet above sea level. Failure to meet the requirements of 3.6.5, 3.6.6, 3.6.7, and 3.7.4 shall constitute failure of this test.

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4.7.5 Temperature shock. The eyepiece shall be subjected to the test of Method 503.1, Procedure I, of MIL-STD-810, except that the high temperature chamber shall be maintained at +68°C, the low temperature chamber shall be maintained at -51°C, and Steps 1, 3, and 5 stabilization periods shall be a minimum of two (2) hours. Immediately after each cold cycle, the eyepiece shall be examined for internal condensation. Evidence of internal condensation or failure to meet the requirements of 3.6.5, 3.6.6,3.6.7, and 3.7.5 shall constitute failure of this test.

4.7.6 Humidity. The eyepiece shall be subjected to the test of Method 507.1, Procedure II, of MIL-STD-810, except that only two (2) continuous 48 hour cycles shall be required. Failure to meet the requirements of 3.6.5, 3.6.6, 3.6.7, and 3.7.6 shall constitute failure of this test.

4.7.7 Immersion. The eyepiece shall be subjected to the test of Method 512.1, Procedure I, of MIL-STD-810, except that the immersion period shall not be less than 30 minutes. Any evidence of water leakage or failure to meet the requirements of 3.6.5, 3.6.6., 3.6.7, and 3.7.7 shall constitute failure of this test.

4.7.8 High intensity shock. The eyepiece shall be subjected to the test Method 516.2, Procedure IV, of MIL-STD-810 in accordance with Figure 516.2-2. The shocks shall be half sine pulses having a time duration of 4 milliseconds, ± 5 percent, and a peak amplitude of 100g's (see 3.10.2), ± 15g's except that the shock pulses applied along the axes horizontal and perpendicular to the optical axis shall be 50g's, ± 15 percent. Failure to meet the requirements of 3.6.5, 3.6.6, 3.6.7, and 3.7.8 shall constitute failure of this test.

4.8 Packaging inspection. The inspection of the preservation-packaging packing and marking shall be in accordance with the quality conformance inspection requirements of the applicable specifications and requirements specified in the contract or purchase order.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2.h).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The eyepiece lens is used for aimed firing at night with the Night Vision Sight, Individual Served Weapon AN/PVS-4 and the Night Vision Sight, Crew Served Weapon AN/TVS-5.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of specification.
b. Quantity and schedule for IPT testing and disposition of IPT samples.
c. Sampling plans for Group B and Group C testing. As guidance, unless otherwise specified, sampling shall be conducted per the requirements of MIL-STD-105.
d. Necessary actions by the contractor in the event of a Group C failure (see 4.5.4).
e. Issue of DODISS to be cited in the solicitation, and if required,the specific issue of individual documents referenced (see 2.1.1 and 2.1.2.)
f. Qualification - If product is not qualified at time of award the contract must require qualification prior to first delivery.
g. MIL-STD-810C shall be used for all environmental tests of section 4 that specify MIL-STD-810.
h. Levels of preservation and packaging (see section 5).

6.3 Definitions. See 3.10

6.4 Qualification. With respect to products requiring qualifications, unless otherwise specified by the contract or purchase order, if a product is not qualified at time of award, the contract shall require qualification prior to first delivery. The activity responsible for approving qualified products is the CECOM, Concurrent Engineering Directorate, ATTN: AMSEL-ED-IN, Fort Monmouth, NJ 07703. Information pertaining to qualification of products may be obtained from the above activity.

6.5 Subject term keyword listing.
Eyepiece
Lens assembly

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - CR

User Activity:
Army - AR

Preparing Activity:
Army - CR

Project 5855-A373

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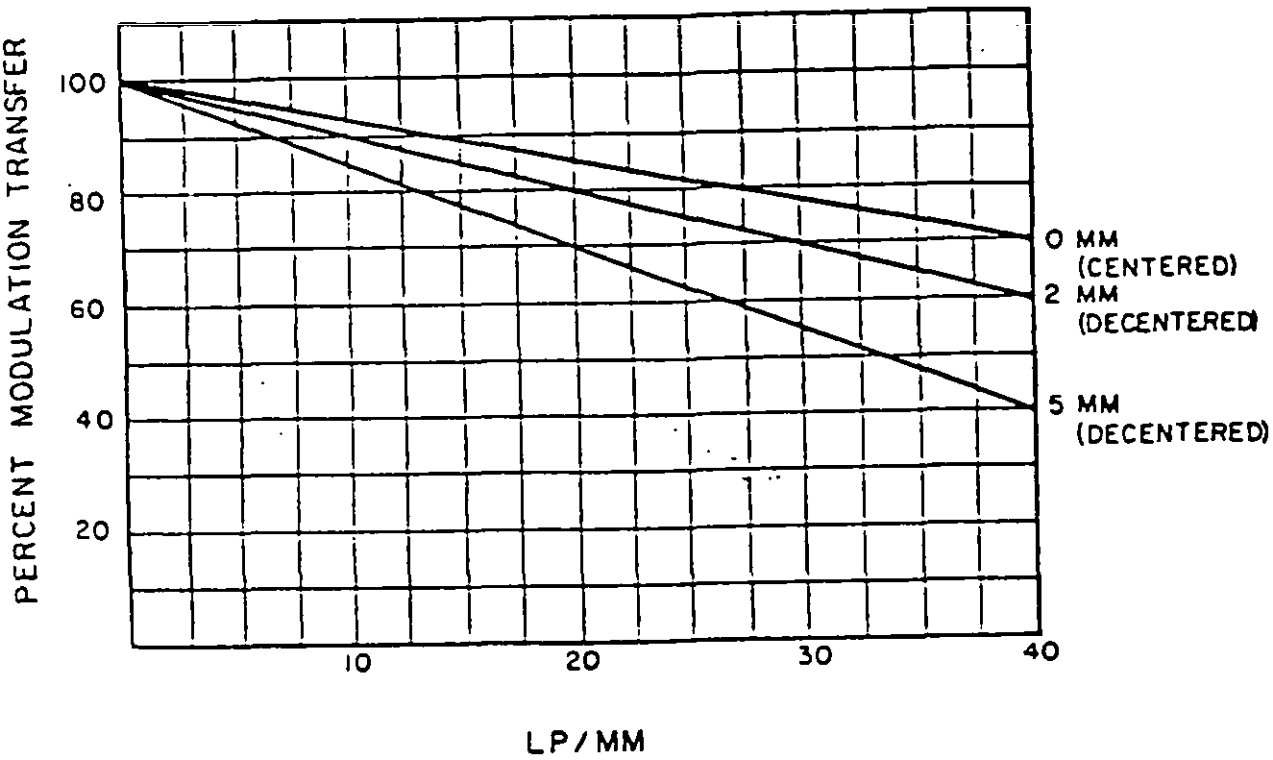


FIG. 1. EYEPIECE MTF ON AXIS

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MODULATION TRANSFER FUNCTION

lp/mm	PINHOLE MTF, %	SLIT MTF, %
0.000	100.000	100.000
2.000	99.500	99.589
4.000	98.000	98.363
6.000	95.600	96.340
8.000	93.000	93.549
10.000	89.000	90.032
12.000	84.000	85.839
14.000	78.500	81.033
16.000	73.000	75.683
18.000	67.000	69.865
20.000	60.000	63.662
22.000	54.000	57.162
24.000	46.900	50.455
26.000	40.000	43.633
28.000	33.000	36.788
30.000	26.000	30.011
32.000	19.000	23.387
34.000	13.500	17.001
36.000	8.000	10.929
38.000	4.000	5.242
40.000	0.000	0.000

PINHOLE DIAMETER - 0.0305MM
SLIT WIDTH - 0.025MM

FIGURE 2

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PINHOLE lp/mm	MTF, %	SLIT MTF, %
0.000	100.000	100.000
2.000	99.500	99.589
4.000	98.000	98.363
6.000	95.600	96.340
8.000	93.000	93.549
10.000	89.000	90.032
12.000	84.000	85.839
14.000	78.500	81.033
16.000	73.000	75.683
18.000	67.000	69.865
20.000	60.000	63.662
22.000	54.000	57.162
24.000	46.900	50.455
26.000	40.000	43.633
28.000	33.000	36.788
30.000	26.000	30.011
32.000	19.000	23.387
34.000	13.500	17.001
36.000	8.000	10.929
38.000	4.000	5.242
40.000	0.000	0.000

PINHOLE DIAMETER - 0.0305MM
SLIT WIDTH - 0.025MM

FIGURE 3. MODULATION TRANSFER FUNCTION

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revisio letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification o requirements on current contracts. Comments submitted on this form do not constitute or imply authorization t waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-L-49368B(CR)	2. DOCUMENT DATE (YYMMDD) 27 May 1992

3. DOCUMENT TITLE
LENS ASSEMBLY, EYEPIECE, AN/PVS-4 AND AN/TVS-5

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME CECOM	b. TELEPHONE (Include Area Code) (1) Commercial (908) 532-5851	(2) AUTOVON 992-5851
c. ADDRESS (Include Zip Code) AMSEL-ED-TM Fort Monmouth, NJ 07703-5023	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340. AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.*

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